

This roadmap is the shared working map for the two-paper split:

- ``thermal_comfort_paper``: thermal comfort paper
- ``ieq_paper``: overall IEQ paper
- ``../02_Datasets``: central location for clean and model-ready Excel datasets
- ``article_figures``: final paper-facing figures used in manuscripts
- ``../04_Figures``: PDF figure mirror used directly by the LaTeX article, split into paper-specific subfolders

The goal is to keep the work reproducible, easy to return to, and clear enough to discuss with supervisors. Every substantial step should update the relevant ``progress.md``, regenerate/export the PDF version, and keep the executable workflow inside one main notebook per paper.

Current Starting Point

- Early exploratory IEQ work is in ``ieq_benchmark/standalone_overall_ieq_models`` and remains historical context only.
- The old five-class Random Forest baselines (``v04``, ``v09``) have been superseded by the current paper-facing three-class workflows in ``thermal_comfort_paper`` and ``ieq_paper``.
- Historical CatBoost work exists in the earlier exploratory benchmark, but newer CatBoost/SVM screens are treated as supplementary notes rather than part of the main paper-facing model line.
- Supporting figures already exist for data completeness, target distributions, model comparisons, and selected supplementary ablation checks.

Completed So Far

Status as of ``2026-05-13``:

- Two-paper workspace created for thermal comfort and overall IEQ.
- Raw Belgian classroom dataset retained unchanged in ``../02_Datasets/raw``.
- Clean datasets created in ``../02_Datasets/clean``:
 - ``thermal_comfort_clean_dataset.xlsx``
 - ``ieq_clean_dataset.xlsx``
- Model-ready imputed datasets created in ``../02_Datasets/model_ready``:
 - ``thermal_comfort_model_dataset.xlsx``
 - ``ieq_model_dataset.xlsx``
- Clean and model-ready datasets are generated by separate scripts:
 - ``_project_tools/create_step1_clean_paper_datasets.py``
 - ``_project_tools/create_step2_imputed_model_datasets.py``
- ``Occupant ID``, ``Group``, ``Subgroup``, and ``Student`` metadata were added to both clean datasets.

- Three-class target columns were added to both clean and model-ready datasets:
- `Thermal satisfaction 3-class`
- `IEQ satisfaction 3-class`
- The primary modeling target for both papers is the three-class satisfaction target; the original five-class target is retained for descriptive figures, traceability, and possible sensitivity checks.
- Imputation strategy was implemented and documented. The model-ready `data` sheets now exclude trace columns and imputation metadata, including `Estimated Age`; detailed imputation counts remain in workbook support sheets.
- Model-ready datasets have no missing values in the defined target/input columns.
- Feature completeness and complete-case availability plots/tables were generated for both papers.
- Paper-facing target distribution figures were generated for:
 - thermal satisfaction, comparing original five-class and unified three-class scales
 - overall IEQ satisfaction, comparing original five-class and unified three-class scales
- Shared figure style guide and plotting helper were added.
- Article-ready figures are collected in `article\_figures` and mirrored as PDFs to `../04\_Figures` where files are not locked.
- Progress logs and roadmap have PDF exports.
- Tuned model-comparison notebooks and output tables exist for both papers:
 - thermal comfort: current strongest tuned model is Random Forest, with accuracy=0.669, macro F1=0.593, and ordinal MAE=0.396.
 - IEQ: current selected single-model line is refined Extra Trees with fixed final hyperparameters.
- For IEQ, a portable supervisor handoff notebook was created:
 - `ieq\_paper/03\_notebook/04\_ieq\_final\_extra\_trees\_supervisor.ipynb`.
- It uses only `../02\_Datasets/model\_ready/ieq\_model\_dataset.xlsx`, fixed final Extra Trees hyperparameters, deterministic 5-fold cross-validation, a confusion-matrix heatmap, and a reference comparison figure.
- After removing `Estimated Age` from the model-ready `data` sheet, the current IEQ handoff run uses 20 predictors and reaches accuracy=0.722, macro F1=0.591, weighted F1=0.719, and ordinal MAE=0.316.
- The same handoff-notebook pattern can be copied to thermal comfort with the tuned Random Forest configuration.

Folder Structure

Each paper has the same structure:

- `00\_progress`: running progress log, meeting notes, PDF exports
- `01\_data\_raw`: paper-specific raw-data notes if needed; central raw files are stored in `../02\_Datasets/raw`
- `02\_data\_clean`: cleaning notes and data-definition notes; datasets are stored centrally in `../02\_Datasets`
- `03\_notebook`: one main reproducible notebook for the paper
- `04\_outputs/figures`: generated figures

- `04\_outputs/tables`: generated tables and metric summaries
- `05\_literature`: article list, citation notes, comparison tables
- `06\_manuscript`: manuscript outline, paragraphs, draft text
- `07\_transfer\_learning`: external datasets and transfer-learning experiments

Working Rules

- Keep one main notebook per paper:
- `thermal\_comfort\_paper/03\_notebook/thermal\_comfort\_workflow.ipynb`
- `ieq\_paper/03\_notebook/ieq\_workflow.ipynb`
- Keep supervisor handoff notebooks separate from exploratory/tuning notebooks. For IEQ, the current handoff notebook is
`ieq\_paper/03\_notebook/04\_ieq\_final\_extra\_trees\_supervisor.ipynb`.
- Keep internal helper scripts in `\_project\_tools`; do not leave one-off maintenance scripts in the project root.
- Use the shared figure style guide for all new paper figures: `PAPER\_FIGURE\_STYLE\_GUIDE.md`.
- Generate Matplotlib figures through `\_project\_tools/paper\_style.py` when practical, so colors, fonts, and exports stay consistent.
- Save manuscript-ready exported figures in `article\_figures`.
- Mirror PDF versions of manuscript-ready figures to the relevant `./04\_Figures` subfolder.
- After every meaningful work session:
- update the paper-specific `00\_progress/progress.md`
- add changed outputs to the roadmap/progress notes
- export the updated Markdown progress to PDF
- Store raw datasets in `./02\_Datasets/raw`, clean datasets in `./02\_Datasets/clean`, and imputed model-ready datasets in `./02\_Datasets/model\_ready`; keep paper folders free of duplicate dataset copies.
- Keep raw data unchanged in `./02\_Datasets/raw`.
- Use English for final figure titles, captions, manuscript paragraphs, and paper-ready tables.
- Use Czech or English for internal notes, whichever is faster and clearer.
- Overall target: finish the work up to and including transfer-learning readiness by `2026-05-25`.
- If a deadline slips by more than 1-2 days, reduce scope before adding new experiments.
- Every Friday: update `progress.md`, export PDFs, and write the next three concrete actions.
- PDF export command: `& .\venv\Scripts\python.exe "\_project\_tools\export\_project\_docs\_to\_pdf.py"`

Step 0: Create the Two-Paper Workspace

Deadline: `2026-05-04` (setup)

Checkpoint: Done when both paper folders, progress logs, starter notebooks, and PDF roadmap exist.

- [x] Create `thermal\_comfort\_paper`.

- [x] Create `ieq\_paper`.
- [x] Add matching folder structure to both papers.
- [x] Add a root roadmap file.
- [x] Add a progress log for each paper.
- [x] Add one main starter notebook for each paper.
- [x] Export this roadmap to PDF.

Step 1: Split the Data Into Two Datasets

Deadline: `2026-05-06` (Week 1)

Checkpoint: On track if both clean Excel files exist. Slightly behind if one dataset is still undefined. Critical delay if target columns are still undecided.

Thermal comfort:

- [x] Define the thermal comfort target variable.
- [x] Define thermal comfort input variables.
- [x] Decide whether to use only sensor/environment variables or also occupant/context variables.
- [x] Create a clean working extract for thermal comfort.
- [x] Save the clean thermal dataset in `../02\_Datasets/clean`.
- [x] Add a short data dictionary.

IEQ:

- [x] Define the IEQ target variable.
- [x] Define the IEQ input variables.
- [x] Decide final physical/context feature sets.
- [x] Create a clean working extract for IEQ.
- [x] Save the clean IEQ dataset in `../02\_Datasets/clean`.
- [x] Add a short data dictionary.

Done means: each paper has one central clean Excel file in `../02\_Datasets/clean`, one data dictionary, and a short note explaining what was kept or excluded.

Step 2: Clean Data and Document Missingness

Deadline: `2026-05-08` (Week 1)

Checkpoint: On track if each paper has missingness figures, cleaning notes, and a clean imputation strategy. Slightly behind if figures exist but paragraphs are missing. Critical delay if missingness strategy is still undecided.

For each paper:

- [x] Make a feature completeness plot.

- [x] Make a complete-case availability plot for the planned model input sets.
- [x] Make a target distribution plot before class simplification.
- [x] Identify variables with high missingness.
- [x] Decide imputation strategy for numeric variables.
- [x] Decide imputation strategy for categorical variables.
- [x] Decide whether to add missing-value indicators.
- [x] Write a paragraph explaining missingness and imputation.
- [x] Save plots to `04\_outputs/figures`.
- [x] Save cleaning notes to `02\_data\_clean/cleaning\_notes.md`.
- [x] Manually check Excel files
- [x] Write a paragraph in IEQ paper
- [x] Write a paragraph in Thermal paper
- [x] Manually checking clearing data code.

Recommended starting rule:

- Numeric values: median imputation.
- Numeric missingness: consider explicit missing-value indicator columns when the feature is important and missingness may carry information, but keep them only if the validation gain is large enough to justify the added feature complexity.
- Categorical values: `Unknown` category.
- Keep complete-case counts visible in internal analysis, but decide later whether paper figures should show them.

Current implemented imputation rule:

- Clean datasets are stored in `../02\_Datasets/clean`; model-ready imputed datasets are generated separately in `../02\_Datasets/model\_ready`.
- `\_project\_tools/create\_step1\_clean\_paper\_datasets.py` creates the clean datasets.
- `\_project\_tools/create\_step2\_imputed\_model\_datasets.py` reads the clean datasets and creates the model-ready datasets.
- `Occupant ID`, `Group`, `Subgroup`, and `Student` metadata were added to both clean datasets.
- `Age` is transferred first from matching `Occupant ID` records, then estimated from `Group` + `Student`, `Group`, and global medians.
- `CLO`, `Ttrend`, `Sound`, and `Lighting` use group-aware median rules with broader fallbacks.
- `Gender` is transferred from `Occupant ID` where possible, otherwise assigned by seeded random draw from the observed group distribution.
- Classroom position is retained as the raw `LocationBack`, `LocationFront`, `LocationLeft`, `LocationRight`, and `LocationMiddle` binary flags instead of being collapsed into one classroom-position category.
- Unknown classroom location is defaulted to `LocationMiddle`.
- The model-ready `data` sheets remove trace/context columns (`source\_row\_number`, `Occupant ID`, `Group`, `Subgroup`) and imputation metadata columns. The `Estimated Age` flag was removed from the paper-facing `data` sheets on 2026-05-13 because it was not needed as a model input; detailed imputation counts remain in the workbook support sheets.

- Broad missingness indicators (`Imputed CLO`, `Imputed Age`, `Imputed Gender`, `Imputed Location`, `Imputed Ttrend`, IEQ-specific `Imputed Lighting`/`Imputed Sound`, and the compact `Estimated Age` flag) were tested and then removed. The fixed-parameter gains were marginal and not robust enough to justify adding these columns to the paper-facing model-ready datasets.
- Class-imbalance strategies were also tested on 2026-05-07. Thermal synthetic sampling, sample-weighted fitting, and probability multiplier calibration did not improve the primary macro F1 result enough to replace the tuned thermal Random Forest baseline. For IEQ, sample weighting and probability multiplier calibration did not beat the refined Extra Trees baseline; earlier IEQ synthetic sampling had also failed to improve macro F1.
- Primary-school-only training was tested on 2026-05-08. It did not improve the best thermal comfort model on primary-school rows. For IEQ, the primary-only Extra Trees model reached accuracy=0.762 and macro F1=0.624, compared with the all-school Extra Trees baseline accuracy=0.725 and macro F1=0.597. The same-population comparison is smaller but still positive: the all-school model evaluated only on primary rows reached accuracy=0.758 and macro F1=0.614. Treat this as a strong IEQ subset/sensitivity result, not as a replacement for the all-school model.

Step 3: Simplify Satisfaction Classes

Deadline: `2026-05-08` (Week 1)

Checkpoint: Done because both original and simplified target distributions are saved and the main target scale is documented.

For each paper:

- [x] Keep the original five-class target available.
- [x] Create a three-class target:
 - `1-2`: dissatisfied
 - `3`: neutral
 - `4-5`: satisfied
- [x] Plot original five-class distribution.
- [x] Plot three-class distribution.
- [x] Record that the main paper result will use the three-class target.
- [x] Document the reason for the chosen target scale.

Decision:

- Both papers will use the unified three-class target as the primary modeling target: dissatisfied (`1-2`), neutral (`3`), and satisfied (`4-5`).
- The original five-class targets remain available for descriptive distributions, traceability to the questionnaire scale, and optional sensitivity analysis.
- This target scale is preferred for the article models because it reduces sparsity in the rare low-vote categories and matches the article-facing interpretation of satisfaction as dissatisfied, neutral, or satisfied.

Step 4: Literature and Citations

Deadline: `2026-05-20` (Week 3; start collecting papers in parallel after Step 2)
Checkpoint: On track if each paper has at least 8-10 useful references in the literature matrix. Slightly behind if references are collected but not structured. Critical delay if no fair benchmark paper is identified.

For each paper:

- [] Collect 10-20 core papers.
- [] Separate literature into:
 - dataset/method papers
 - thermal comfort or IEQ prediction papers
 - model comparison papers
 - transfer-learning or external validation papers
- [] Create a literature matrix in `05\_literature/literature\_matrix.md`.
- [] Record target scale used by each article.
- [] Record input features used by each article.
- [] Record models and reported accuracy/metrics.
- [] Identify the closest benchmark paper.
- [] Write a short comparison paragraph.

Done means: each paper has a citation table and a clear statement of what result is a fair comparison.

Step 5: Build Baseline Models

Deadline: `2026-05-11` (Week 1)
Checkpoint: On track if both papers have first comparable model comparison tables. Slightly behind if only Random Forest is complete. Critical delay if one paper still has no runnable model pipeline.
Scope rule: if Step 5 slips, finish Random Forest and one simple baseline first; do not expand the model list until both paper pipelines run end-to-end.

For each paper, train and compare:

- Primary modeling targets:
 - thermal comfort paper: `Thermal satisfaction 3-class`
 - IEQ paper: `IEQ satisfaction 3-class`
- [x] Logistic/ordinal regression baseline.
- [x] Random Forest.
- [x] XGBoost or gradient boosting.
- [x] ANN / MLP.

For each model:

- [x] Use the same train/test or cross-validation scheme.
- [x] Record exact match accuracy for classification.
- [x] Record balanced accuracy for imbalanced classification.
- [x] Record macro F1.
- [x] Record MAE for ordinal interpretation.
- [x] Save fold-level metrics.
- [x] Save confusion matrices.
- [x] Save prediction distributions.

Current status:

- Thermal comfort has runnable 5-fold comparison and tuned comparison notebooks in ``thermal_comfort_paper/03_notebook``, with output tables in ``thermal_comfort_paper/04_outputs/tables``.
- IEQ has runnable 5-fold comparison and tuned comparison notebooks in ``ieq_paper/03_notebook``, with output tables in ``ieq_paper/04_outputs/tables``.
- The IEQ supervisor handoff notebook now provides a cleaner final-model-only version without hyperparameter search; the same template should be copied for thermal comfort.

Current evaluation standard:

- 5-fold stratified cross-validation.
- Report exact accuracy, balanced accuracy, macro F1, weighted F1, MAE, and per-class recall.
- Keep the same rows and feature set when comparing model families.

Step 6: Hyperparameter Tuning

Deadline: ``2026-05-13`` (Week 2)

Checkpoint: On track if all major model families have at least one tuned variant. Slightly behind if ANN/MLP tuning is missing but RF and boosting are complete. Critical delay if results are not comparable because rows/features differ unexpectedly.

Scope rule: do not tune endlessly. Once one model is clearly defensible, move to model selection and interpretation.

For each paper:

- [x] Define a small conservative tuning grid for each model.
- [x] Tune Random Forest depth, leaf size, feature sampling, and class weights.
- [x] Tune XGBoost/gradient boosting learning rate, depth, estimators, regularization, and class weighting if needed.
- [x] Tune ANN architecture, regularization, learning rate, and early stopping.
- [x] Keep a table of all model variants.
- [x] Record why each variant was tried.
- [x] Avoid choosing a model only by one metric if minority classes collapse.

Done means: each paper has a model comparison table that is supervisor-ready.

Current status:

- Thermal comfort tuning is complete enough for supervisor discussion. The current best model is tuned Random Forest, and rejected ablations are documented in ``thermal_comfort_paper/06_manuscript/modeling_results_notes.md``.
- IEQ tuning is complete enough for supervisor discussion. The current selected single-model configuration is refined Extra Trees: ``n_estimators=700`, `criterion="entropy", `bootstrap=True`, `max_samples=0.9`, `max_features=None`, `max_depth=20`, `min_samples_leaf=2`, `min_samples_split=10`, and `class_weight="balanced"``.
- The IEQ final handoff notebook uses the selected hyperparameters directly instead of running ``RandomizedSearchCV``, so repeated supervisor runs are intended to stay aligned with the article-facing model definition.

Step 7: Select the Best Model

Deadline: ``2026-05-15`` (Week 2)

Checkpoint: On track if each paper has one chosen main model and one backup model. Slightly behind if the best model is selected but figures are missing. Critical delay if the main metric is still undecided.

For each paper:

- [x] Choose primary metric.
- [x] Choose secondary metrics.
- [x] Decide whether the best model should prioritize exact accuracy or minority-class behavior.
- [x] Compare top models visually.
- [x] Select one main model and one defensible alternative.
- [x] Write a short model selection paragraph.

Recommended decision framing:

- Main model: best balance of accuracy and scientific interpretability.
- Alternative model: best minority-class behavior or strongest external comparability.

Current selections:

- Thermal comfort: tuned Random Forest is the current main model by macro F1 and robustness. Extra Trees remains the closest alternative.
- IEQ: refined Extra Trees is the current main single-model result. Tuned Random Forest is the main simple alternative, while the 90/10 Extra Trees + Random Forest soft-voting ensemble is documented as a small-gain extension rather than the primary paper model.
- IEQ now has a final supervisor handoff notebook with the selected Extra Trees model, fixed hyperparameters, a confusion-matrix visualization, and a grey reference comparison against

the earlier model-family results.

- Thermal comfort still needs the same clean supervisor handoff notebook format. This is implementation work rather than methodological work, because the tuned model and comparison outputs already exist.

Step 8: Feature Importance and Interpretation

Deadline: `2026-05-18` (Week 2)

Checkpoint: On track if both papers have feature-importance figures and interpretation notes. Slightly behind if only one paper has interpretation. Critical delay if feature importance contradicts model setup and needs reruns.

For each paper:

- [] Generate feature importance for the selected best model.
- [] If possible, compare feature importance across at least two model variants.
- [] Keep encoded indicators separate when that helps interpretation.
- [] Group indicators only when the paper narrative needs source-level interpretation.
- [] Check whether feature importance changes across folds or hyperparameters.
- [] Write an interpretation paragraph.

Important note:

Feature importance means model reliance, not causal influence. The manuscript should avoid causal wording unless a separate causal design supports it.

Current status:

- This step is still the main next analysis gap for both papers. The selected models exist, but final feature-importance figures and interpretation paragraphs have not yet been produced in the clean supervisor handoff format.
- IEQ should be handled first using the fixed Extra Trees pipeline from ``04_ieq_final_extra_trees_supervisor.ipynb``; thermal comfort can then reuse the same structure with the selected Random Forest model.

Step 9: External Dataset Search

Deadline: `2026-05-22` (Week 3)

Checkpoint: On track if each paper has a feasible external dataset candidate or a justified reason why not. Slightly behind if datasets are found but feature overlap is not checked. Critical delay if transfer-learning feasibility is still unknown.

Scope rule: prioritize dataset compatibility and feasibility over complex transfer-learning methods.

- [x] Contact authors of norwegian dataset

Thermal comfort:

- [] Identify public thermal comfort datasets.
- [] Check target compatibility.
- [] Check feature overlap.
- [] Check whether classroom/student/Covid context is comparable.
- [] Create an external dataset compatibility table.

IEQ:

- [] Identify public IEQ or indoor-environment satisfaction datasets.
- [] Check whether overall IEQ or domain satisfaction is available.
- [] Check target scale compatibility.
- [] Check feature overlap.
- [] Create an external dataset compatibility table.

Done means: each paper has at least one candidate external dataset and a written note explaining whether transfer is feasible.

Step 10: Transfer Learning / External Validation

Deadline: `2026-05-25` (Week 3 final milestone)

Checkpoint: On track if each paper has either a first transfer result or a clear transfer feasibility conclusion. Slightly behind if only one paper has transfer status. Critical delay if external dataset search is unresolved.

For each paper:

- [] Start with direct external validation if feature overlap allows it.
- [] Compare model performance on internal CV vs external data.
- [] Try feature alignment and common-feature-only models.
- [] Try target-scale harmonization if needed.
- [] Try fine-tuning only if the external dataset has enough compatible labels.
- [] Document what transfers and what fails.

Possible outcomes:

- Strong transfer: model generalizes reasonably well.
- Partial transfer: only some feature groups or target classes transfer.
- Weak transfer: dataset is too context-specific, which itself can be a publishable finding.

Progress Tracking Template

Use this after each work session in the relevant `progress.md`:

Done

Decisions

Results

Open Questions

Next Actions

Immediate Next Actions

- [x] Source the raw dataset for the thermal comfort paper.
- [x] Source the raw dataset for the IEQ paper.
- [x] Define exact target columns for both papers.
- [x] Define first-pass feature lists for both papers.
- [x] Generate the two clean Excel files.
- [x] Add first missingness/completeness plots.
- [x] Add target distribution plots and the first imputation paragraph.
- [x] Generate model-ready datasets without imputation metadata columns in the paper-facing `data` sheets.
- [x] Build and verify the IEQ final supervisor handoff notebook with fixed Extra Trees hyperparameters.
- [x] Add IEQ confusion-matrix and reference model-comparison visualizations to the handoff notebook.
- [] Create the matching thermal comfort supervisor handoff notebook using the selected tuned Random Forest model.
- [] Add final feature-importance figures for IEQ Extra Trees.
- [] Add final feature-importance figures for thermal comfort Random Forest.
- [] Update manuscript result notes so IEQ values consistently reflect the current 20-predictor dataset without `Estimated Age`.